Relativistic Statistical Mechanics

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1 Transformation Laws Table

The following table (see table 1) shows the results achieved for some authors [1], [2], [3], [4], [5] and the results of the authors of this document for the transformation laws of the thermodynamic variables. It is shown the agreement of this work with [5] and the partial agreement with [3] and [4].

2 The invariance of entropy

Theorem 1. The entropy $S = k_B \ln \Omega$ for a moving inertial observer, K,with respect to the container of an ideal gas (N equal particles of same mass, elastic collisions between the particles and the walls, and non-interacting particles) is equal to the entropy $S_0 = k_B \ln \Omega_0$ for a rest observer with the container, K_0.

Where $\Omega = \Omega(E, V, N)$ is the number of states function for K depending on internal energy E, volume V, and number of particles N. Analogously, $\Omega_0 = \Omega_0(E_0, V_0, N_0)$ is the number of states function for K_0.

Proof:
By definition:

$$\Omega = \frac{1}{N!}\frac{1}{h^{3N}} \underbrace{\int_{\mathcal{V}} d^3r_1 \cdots d^3r_N}_{\Omega_v} \underbrace{\int_{R} d^3p_1 \cdots d^3p_N}_{\Omega_p}, \tag{1}$$

where h is the planck constant, and $\vec{r}_i$ and $\vec{p}_i$ are the ith position and momentum for K. Also, $\mathcal{V}$ is the volume space, R is the momentum space defined as:

$$R = \left\{ (\vec{p}_1, ..., \vec{p}_N) \mid \sum_{i=1}^{N} c\sqrt{m^2c^2 + |\vec{p}_i|^2} = E \right\}. \tag{2}$$

	Einstein, Planck	**Ott, Arzelièz**	**Sutcliffe**	**Guarín-Carvajal-Páez-Gutiérrez**
Pressure	$P = P_0$		$P_m = P_0$, $P_{th} = P_0\gamma^2$	$P = P_0\gamma^2$
Volume	$V = V_0/\gamma$			
Entropy	$S = S_0$			
Temperature	$T = T_0/\gamma$	$T = T_0\gamma$		
Heat	$\delta Q = \delta Q_0/\gamma$	$\delta Q = \delta Q_0\gamma$		
Internal energy	$E = E_0\gamma + P_0V_0\gamma v^2/c^2$	$E = E_0\gamma$		
Helmholtz free energy	$F = F_0/\gamma$	$F = F_0\gamma$		

Table 1: Transformation of the thermodynamic quantities for different authors, adapted from [6]. A zero subindex indicates variables measured for a rest observer with the container of the gas. By convention $\gamma = 1/\sqrt{1 - v^2/c^2}$, where v is the velocity of the moving observer respecting to the rest observer, P_m is the *mechanical pressure* and P_{th} the *thermodynamic* one (as discussed by [5]).

By the non-interacting condition, the position $\vec{r}_i$ of each particle is independent one of the other. And therefore we can separate Ω_v as a product of integrals:

$$\Omega_v = \left(\int_V d^3r_1\right)\cdots\left(\int_V d^3r_N\right) = \left(\int_V d^3r_1\right)^N = V^N\,. \tag{3}$$

Lemma 1. The volume V_0 for the observer K_0 changed to $V_0/\gamma = V$ by a Lorentz transformation corresponding to a Lorentz Boost in one direction. γ is the Lorentz factor.

By the previous lemma:

$$\Omega_v = \frac{V^N}{\gamma^N} = \frac{\Omega_{v0}}{\gamma^N}\,. \tag{4}$$

By the other side, chosen an appropriate coordinate axis we can say that the Lorentz Boost occur in the $x-$axis. According to [7] the transformation laws for momentum in this case are:

$$p_x = \gamma\left(g_x + \frac{v}{c^2}E_0\right)\,, \tag{5}$$

$$p_y = g_y\,, \tag{6}$$

$$p_z = g_z\,, \tag{7}$$

where $\vec{p}$ is the momentum of one particle for observer K, and $\vec{g}$ is the momentum of the same particle but for K_0. Besides, v is the relative speed between the observers, $\gamma = 1/\sqrt{1 - v^2/c^2}$, c is the speed of light, and $E_0 = c\sqrt{m^2c^2 + g^2} = mc^2\sqrt{1 + \frac{g^2}{m^2c^2}}$ is the relativistic energy of the particle for K_0.

We want to implement a change of variables from p_i to g_i in the momentum integrals. The Jacobian of the transformation is:

$$J(\vec{g}) = \frac{\partial p_{x^i}}{\partial g_{x^i}} = \begin{vmatrix} \frac{\partial p_x}{\partial g_x} & \frac{\partial p_x}{\partial g_y} & \frac{\partial p_x}{\partial g_z} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{\partial p_x}{\partial g_x} = \frac{\partial}{\partial g_x}\left[\gamma\left(g_x + \frac{v}{c^2}mc^2\sqrt{1 + \frac{g^2}{m^2c^2}}\right)\right]\,. \tag{8}$$

	Einstein, Planck	**Ott, Arzelièz**	**Sutcliffe**	**Guarín-Carvajal-Páez-Gutiérrez**
Pressure	$P = P_0$		$P_m = P_0$, $P_{th} = P_0\gamma^2$	$P = P_0\gamma^2$
Volume	$V = V_0/\gamma$			
Entropy	$S = S_0$			
Temperature	$T = T_0/\gamma$	$T = T_0\gamma$		
Heat	$\delta Q = \delta Q_0/\gamma$	$\delta Q = \delta Q_0\gamma$		
Internal energy	$E = E_0\gamma + P_0V_0\gamma v^2/c^2$	$E = E_0\gamma$		
Helmholtz free energy	$F = F_0/\gamma$	$F = F_0\gamma$		

Table 2: Transformation of the thermodynamic quantities for different authors, adapted from [6]. A zero subindex indicates variables measured for a rest observer with the container of the gas. By convention $\gamma = 1/\sqrt{1 - v^2/c^2}$, where v is the velocity of the moving observer respecting to the rest observer, P_m is the *mechanical pressure* and P_{th} the *thermodynamic* one (as discussed by [5]).

Knowing that $g^2 = {g_x}^2 + {g_y}^2 + {g_z}^2$:

$$J(\vec{g}) = \gamma \left(1 + \frac{v}{c^2} mc^2 \frac{g_x/(m^2c^2)}{\sqrt{1 + \frac{g^2}{m^2c^2}}} \right) = \gamma \left(1 + \frac{v}{c^2} \frac{g_x/m}{\sqrt{1 + \frac{g^2}{m^2c^2}}} \right) . \tag{9}$$

Then, by implement this coordinate transformation for all the variables of the N particles:

$$\Omega_p = \int_R d^3p_1 \cdots d^3p_N = \int_R \gamma^N \prod_{i=1}^{N} \left(1 + \frac{v}{c^2} \frac{g_{x_i}/m}{\sqrt{1 + \frac{{g_i}^2}{m^2c^2}}} \right) d^3g_1 \cdots d^3g_N \, . \tag{10}$$

By using equations 4 and 10 we have that the number of states function is:

$$\Omega = \frac{1}{N!} \frac{1}{h^{3N}} \frac{V^N}{\gamma^N} \int_R \gamma^N \prod_{i=1}^{N} \left(1 + \frac{v}{c^2} \frac{g_{x_i}/m}{\sqrt{1 + \frac{{g_i}^2}{m^2c^2}}} \right) d^3g_1 \cdots d^3g_N \, , \tag{11}$$

$$\Omega = \frac{1}{N!} \frac{1}{h^{3N}} V^N \int_R \gamma^N \prod_{i=1}^{N} \left(1 + \frac{v}{c^2} \frac{g_{x_i}/m}{\sqrt{1 + \frac{{g_i}^2}{m^2c^2}}} \right) d^3g_1 \cdots d^3g_N \tag{12}$$

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